**25901399** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Core Course (DSC) - MG4DSCMBG202 - APPLIED MICROBIOLOGY

(2024 ADMISSION ONWARDS)

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. List out fermented milk products. [K] / [CO1]
2. Name the enzyme is involved in nitrogen fixation? [K] / [CO2]
3. Name two commonly occuring bacterial genera in marine water. [K] / [CO3]
4. Write a note on miso. [K] / [CO1]
5. Name two fungi that can be used as SCP? [K] / [CO1]
6. Name the biofertilizer which helps in fixing atmospheric nitrogen in leguminous plants. [K] / [CO2]
7. What is a bioinsecticide? [K] / [CO2]
8. What is littoral zone. [K] / [CO3]
9. Name two useful microbes in food production. [K] / [CO1]

10. Name a protozoan disease that can be transmitted through contaminated water. [K] / [CO3]

11. Give examples for foodborne illness. [K] / [CO1]

12. Give examples for airborne fungal infections. [K] / [CO4]

**[2x10 = 20]**

## Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain mutualism in soil microorganisms. [U] / [CO2]

14. Explain the role of microbes in beer production. [U] / [CO1]

15. Define plankton.Explain phytoplanktons and zooplanktons with examples. [U] / [CO3]

16. Describe the advantages and limitations of microbial biopesticides. [U] / [CO2]

17. Explain different impaction method of air sampling. [U] / [CO4]

18. Describe the Most Probable Number (MPN) method for detecting coliforms in milk. [U] / [CO1]

**[5x4 = 20]**

## Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Explain the different types of spoilage of milk. [K] / [CO1]

20. Explain the zonation of lakes with a neat labeled diagram and describe the microbial communities in lakes. [U] / [CO3]

**[10x1 = 10]**